

Why Likelihood Intervals Work: A Graphical Guide

Reference notes on the score, Fisher information, and the sampling distribution of the MLE

These notes explain why the standard error of a maximum likelihood estimate is $1/\sqrt{\text{curvature}}$, why the MLE is approximately normal, and where the 1.92 cutoff for likelihood ratio intervals comes from. The approach is graphical; the only math needed is a derivative and the central limit theorem.

The running example is the pika survey from lecture: active haypiles counted in n talus plots, each count Poisson with mean λ . To see how estimates behave across repeated surveys, we simulate surveys from a known truth, $\lambda_0 = 3$.

1. Two views of the same curve

For one survey, the negative log-likelihood (NLL) is a valley whose bottom is the MLE, $\hat{\lambda}$. Its slope, with the sign flipped so that it points uphill on the log-likelihood, is called the **score**:

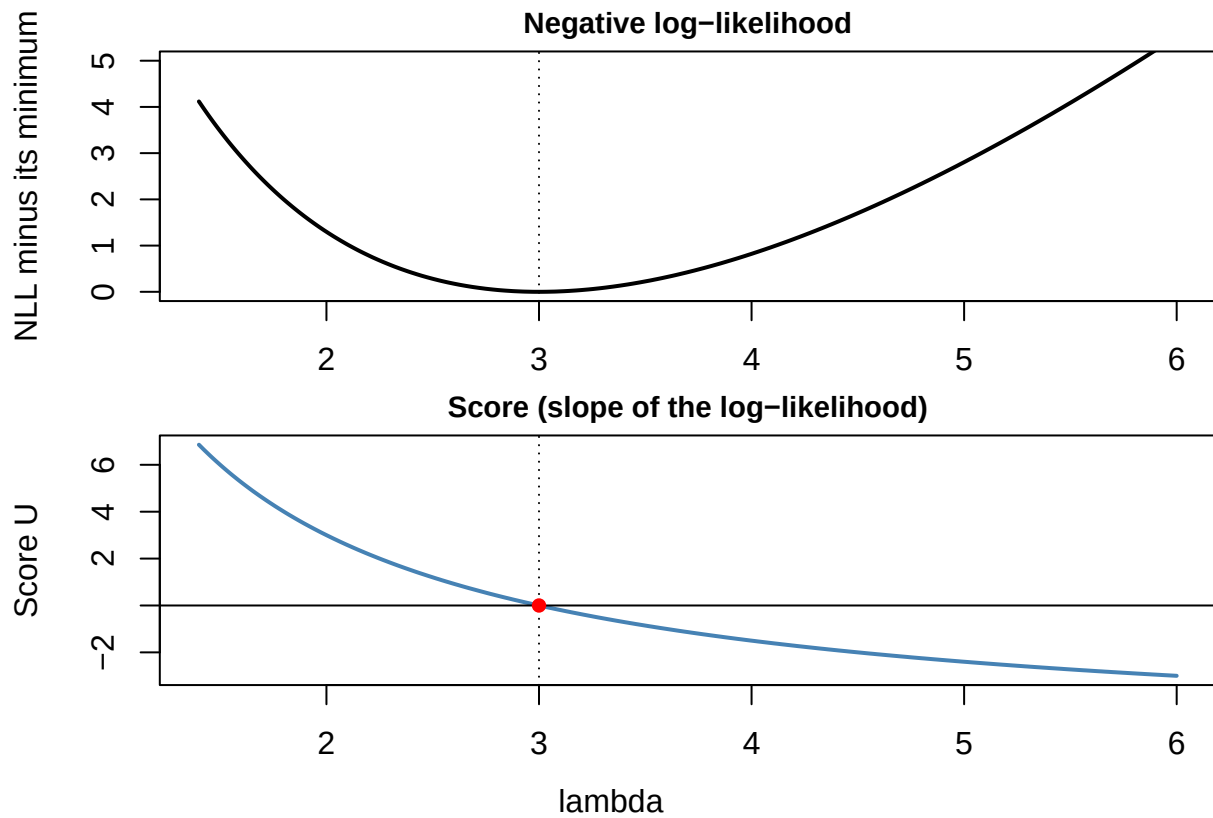
$$U(\lambda) = \frac{d}{d\lambda} \log \mathcal{L}(\lambda) = -\frac{d}{d\lambda} \text{NLL}(\lambda).$$

For the Poisson, $U(\lambda) = \sum y_i/\lambda - n$.

For the data 2, 5, 3, 0, 4, 4 (figure below):

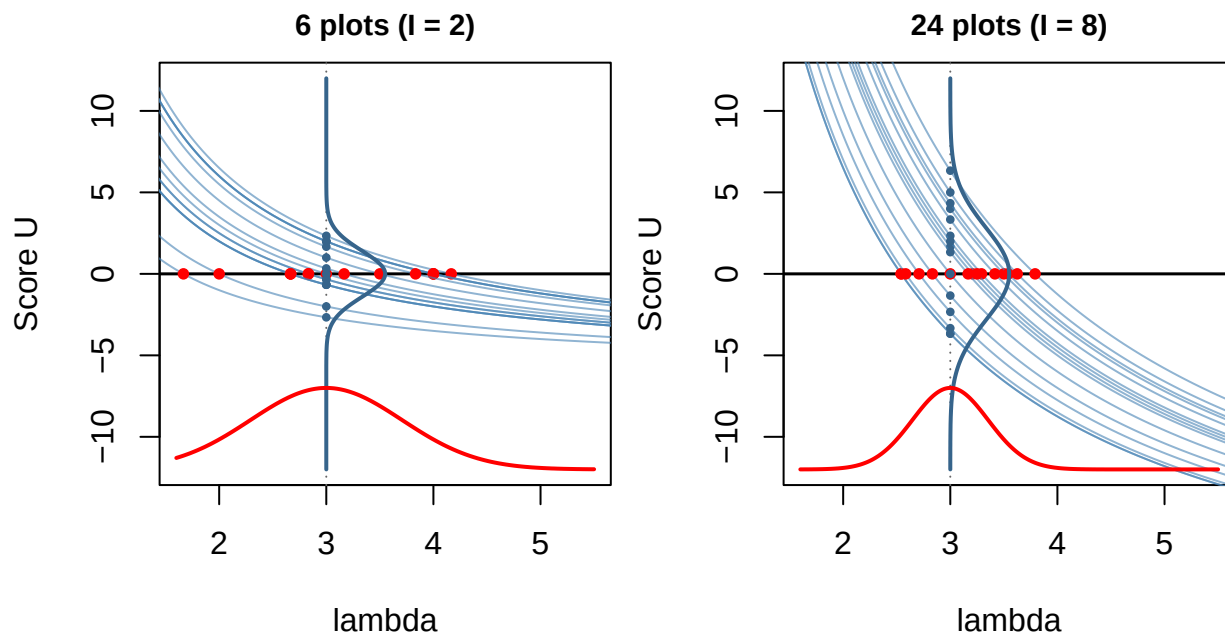
- The bottom of the valley (top panel) is where the score crosses zero (bottom panel). Both mark the MLE, $\hat{\lambda} = 3$.
- The **curvature** of the valley is the **steepness of the score**. A sharp valley means a steep score line; a broad valley means a shallow one.
- Near the MLE, a **parabola-shaped valley** and a **straight-line score** are the same statement. Everything below depends on it.

The expected steepness of the score (the expected curvature of the NLL) is the **Fisher information**, I . For the Poisson, $I = n/\lambda$.



2. Many surveys, many score curves

A new survey gives new counts, so a new score curve. The figure shows the score from 15 simulated surveys at two sample sizes.

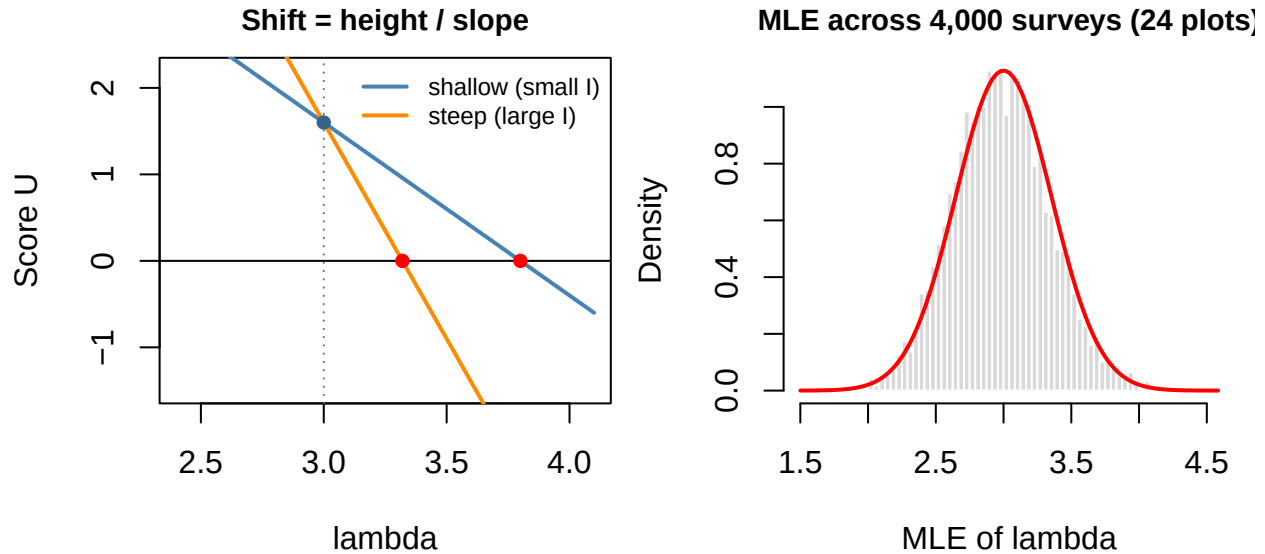


Read it in four steps.

1. **Each curve crosses zero at that survey's MLE** (red dots). The crossings spread along the axis: that spread is the sampling distribution of $\hat{\lambda}$.
2. **At the true value, look up and down.** The blue dots are each curve's height at $\lambda_0 = 3$. They are centered on 0, because at the truth the log-likelihood is flat on average.
3. **Those heights are normal.** The score is a sum over plots, $U(\lambda_0) = \sum_i (y_i/\lambda_0 - 1)$, so the central limit theorem applies. Their variance turns out to be I (section 4 explains why), drawn as the blue bell.
4. **Each curve slopes down at about $-I$.** Section 3 turns this into the spread of the red dots.

3. Turning vertical spread into horizontal spread

Zoom in near the truth, where each score curve is close to a straight line with slope $-I$.



- **Left:** a curve that sits a height h above zero at the truth has to travel h/I to the right before it reaches zero. A steeper line (larger I) gets there sooner, so its MLE lands closer to the truth.
- The heights have variance I . Dividing by the constant slope I gives the MLEs a variance of

$$\frac{\text{Var}(U)}{I^2} = \frac{I}{I^2} = \frac{1}{I}.$$

- **Right:** across 4,000 simulated surveys of 24 plots, the MLEs follow the predicted $\text{Normal}(\lambda_0, 1/I) = \text{Normal}(3, 3/24)$ (red curve).

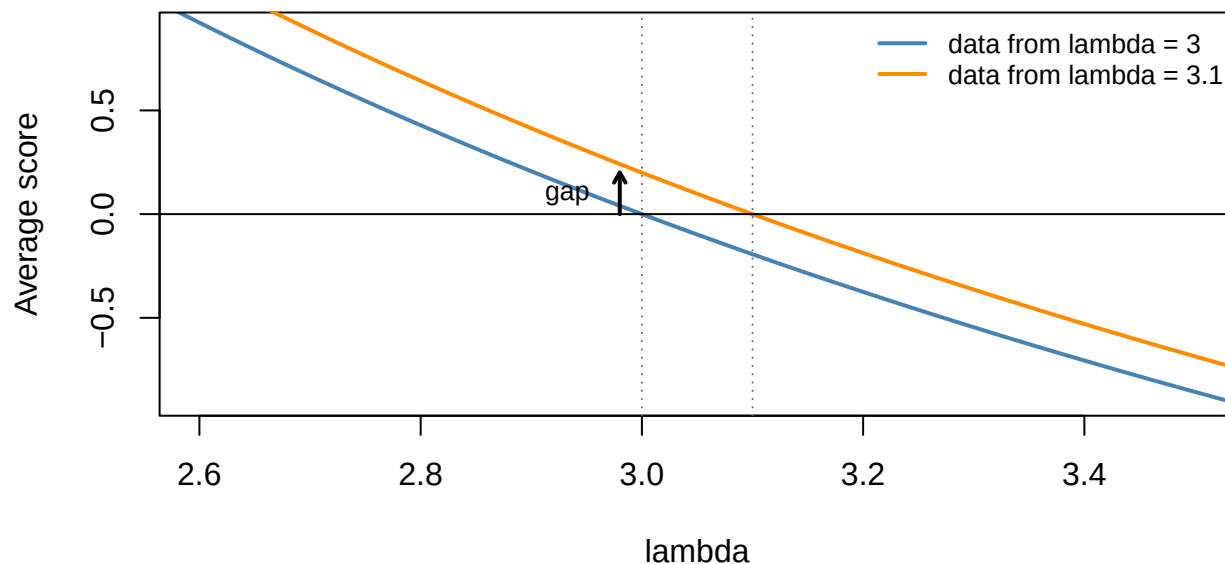
More data. With 24 plots instead of 6, the heights at the truth spread *more* (their variance I grows with n), but the lines get steeper just as fast (their slope is also I). Spread grows like $\sqrt{n}$, slope like n , so the MLEs tighten like $1/\sqrt{n}$. This also settles a common puzzle: more data lowers the maximum likelihood itself, but what matters for the estimate is the steepness, not the height.

Check with the pikas. Here the straight-line step is exact:

$$\hat{\lambda} - \lambda_0 = \frac{U(\lambda_0)}{n/\lambda_0} = \frac{\sum y_i/\lambda_0 - n}{n/\lambda_0} = \bar{y} - \lambda_0, \quad \text{Var}(\hat{\lambda}) = \frac{\lambda_0}{n} = \frac{1}{I}.$$

4. Why the variance of the score equals its slope

Section 3 needed the variance of the score and its slope to be the same number, I . This is the **information identity**, and the score curves show why it has to hold: **the curves must move when the truth moves.**



The blue curve is the average score when the data come from $\lambda_0 = 3$; it crosses zero at 3. Now nudge the truth by $\delta = 0.1$, to 3.1 (orange). The new curve must cross zero at 3.1, or the MLE wouldn't track the truth. The **gap** is how far the average score at $\lambda = 3$ rises. There are two ways to get it.

Route 1: what the slope requires. The curves fall at slope $-I$, so to cross zero δ further right, the new curve must sit about $\delta \cdot I$ above zero at 3. For six plots, $I = 6/3 = 2$, so the gap should be $0.1 \times 2 = 0.2$.

Route 2: what reweighting the surveys delivers. Keep reading the score at $\lambda = 3$. For six plots with total count S , that's $U(3) = S/3 - 6$, a fixed formula applied to each survey. Changing the truth doesn't change the formula; it changes **how often each survey turns up**, and the average is a probability-weighted sum over surveys.

How much more likely does a survey get? The score is, by definition, how fast a survey's log-probability rises as λ increases. So a small step δ multiplies its probability by about $e^{\delta U} \approx 1 + \delta U$:

Total count S	Score at 3, U	Exact ratio	$1 + 0.1 U$
12	-2	0.813	0.8
18	0	0.990	1.0
24	2	1.206	1.2

Surveys whose score points toward a larger λ become more common, and surveys pointing lower become rarer. The new average score at 3 is then

$$\sum U \cdot P_3 \cdot (1 + \delta U) = \underbrace{E[U]}_0 + \delta E[U^2] = \delta \cdot \text{Var}(U).$$

The variance appears because reweighting only moves the average when surveys disagree. If every survey had a score of 0, tilting the weights would change nothing; the more the scores spread, the more the average moves. For six plots, $\text{Var}(U) = \text{Var}(S)/9 = 18/9 = 2$, so the gap is again $0.1 \times 2 = 0.2$.

Both routes describe the same gap, so $\delta \cdot I = \delta \cdot \text{Var}(U)$, which means $\text{Var}(U) = I$. In the figure, the gap is 0.2.

The calculus version, for one observation. Write the score as $u = f'/f$, so $f' = uf$. The score averages to zero for every value of the parameter, $E[u] = \int f' dy = \frac{d}{d\theta} \int f dy = 0$. Differentiating that once more:

$$0 = \int (u'f + uf') dy = E[u'] + E[u^2] \Rightarrow E[-u'] = E[u^2] = \text{Var}(u).$$

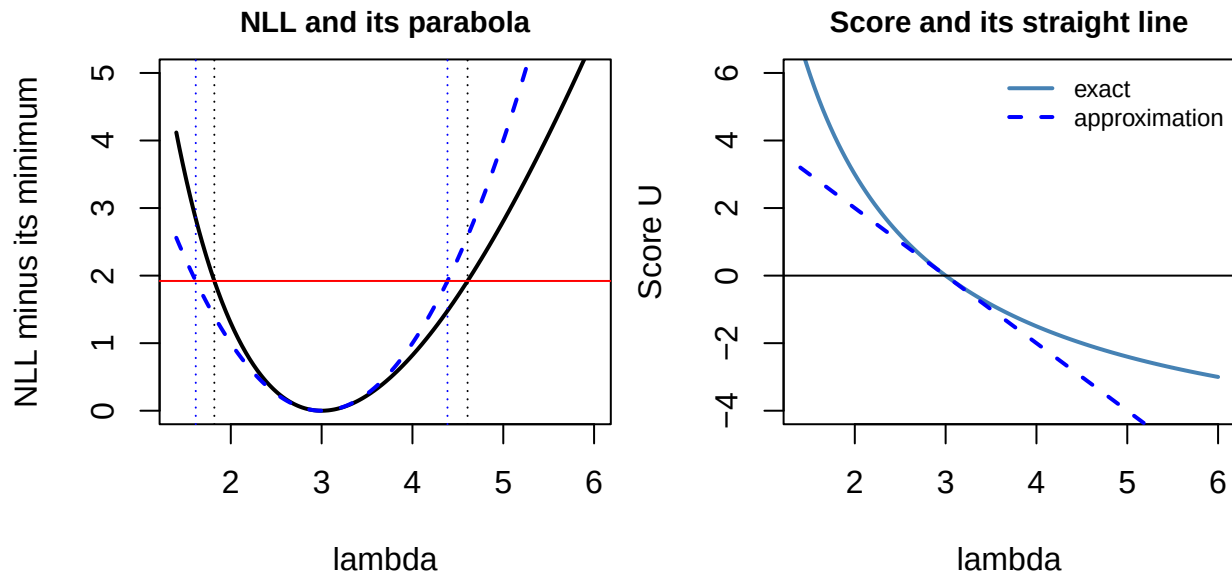
The first term is route 1 (the curve's slope); the second is route 2 (the data reweighted toward high-score outcomes, since $f' = uf$ is the δU in $1 + \delta U$).

5. The parabola, the interval, and the 1.92 cutoff

Sections 2 and 3 described many surveys. In practice we have one, and we use the shape of its NLL near the MLE. The second-order Taylor expansion around $\hat{\lambda}$ (the first derivative is zero there) is

$$\text{NLL}(\lambda) \approx \text{NLL}(\hat{\lambda}) + \frac{1}{2} I (\lambda - \hat{\lambda})^2,$$

a parabola whose curvature is the slope of the score.



- **The interval from the parabola.** Setting the rise equal to the cutoff, $\frac{1}{2} I d^2 = 1.92$, gives $d = 1.96/\sqrt{I}$: the Hessian (Wald) interval, $\hat{\lambda} \pm 1.96 \text{ SE}$, here $[1.61, 4.39]$.

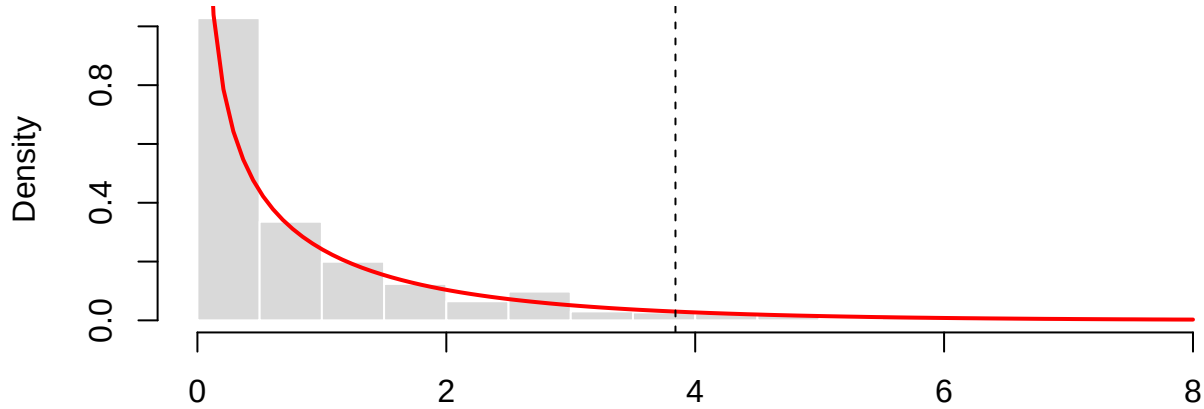
- **Where 1.92 comes from.** Put the true value into the parabola:

$$2 [\text{NLL}(\lambda_0) - \text{NLL}(\hat{\lambda})] \approx I (\hat{\lambda} - \lambda_0)^2 = \left(\frac{\hat{\lambda} - \lambda_0}{1/\sqrt{I}} \right)^2 = Z^2.$$

Section 3 showed that the quantity in parentheses is a standard normal, Z , so twice the NLL difference is χ_1^2 (Wilks' theorem). Its 95th percentile is 3.84, and half of that is 1.92.

- **The likelihood ratio interval** uses the 1.92 cutoff on the true curve instead of the parabola: [1.82, 4.61]. It follows the curve's lopsidedness, which the parabola can't.

So the Hessian interval and the likelihood ratio interval come from the same idea. They agree when the score is close to a straight line, and they part ways when it bends, as it does here with only 6 plots.

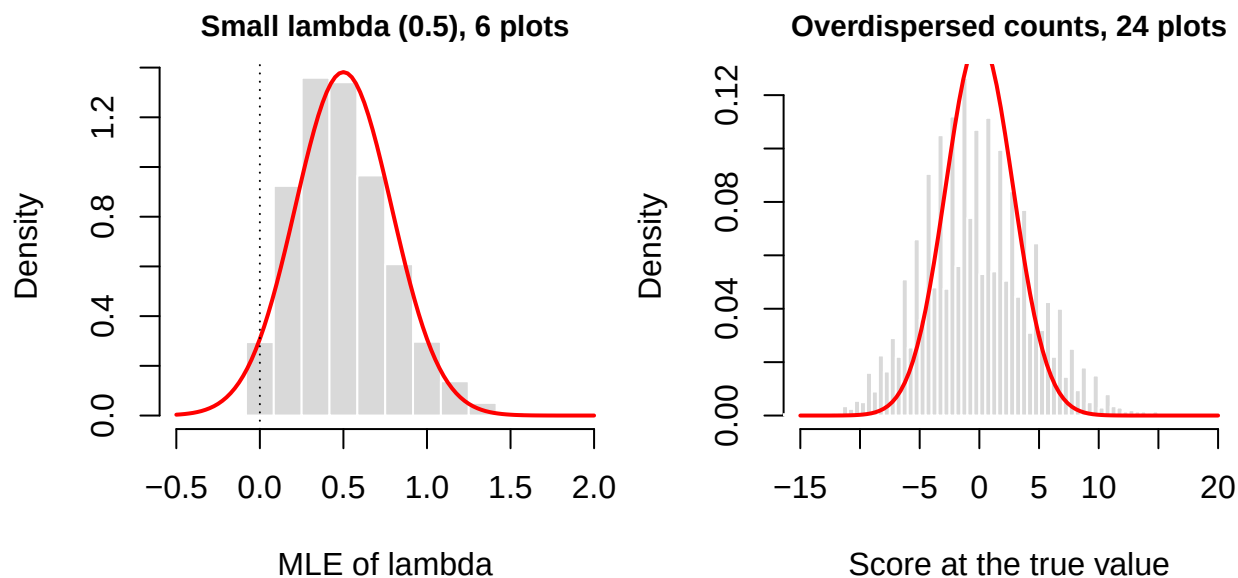


2 x [NLL(true value) – NLL(MLE)], 24 plots

Across 4,000 simulated surveys, twice the NLL difference at the true value follows a χ_1^2 distribution (red curve); about 95% of surveys fall below 3.84 (dashed line). In this simulation the share is 0.948.

6. When the picture breaks

Every step above assumed the score curves are close to straight lines, their heights are close to normal, and the model is right. Each can fail.



- **Near a boundary or with little data (left).** With $\lambda_0 = 0.5$ and 6 plots, the MLE is skewed, and the normal approximation (red) puts probability below zero, where λ can't go. This is why a Hessian interval for a prevalence of 2 in 20 can have a negative lower limit. The likelihood ratio interval or a Bayesian interval handles this better.
- **Wrong model (right).** Here the counts are more variable than a Poisson allows (negative binomial), but we still fit a Poisson. The score heights spread far more than the predicted bell (red). Variance and slope no longer match, so the Hessian standard errors are too small. “Sandwich” (robust) standard errors estimate the two pieces separately for this reason.
- **Several parameters.** The same argument runs with vectors and matrices: $\hat{\theta} - \theta_0 \approx I^{-1}U$, so the variance-covariance matrix of the estimates is I^{-1} , the inverse of the Hessian. Its off-diagonal entries are the correlations between estimates, the tilt of the likelihood surface.

Summary

Idea	Graphical meaning	Formula
Score	slope of the log-likelihood	$U(\theta) = \frac{d}{d\theta} \log \mathcal{L}(\theta)$
MLE	where the score crosses zero	$U(\hat{\theta}) = 0$
Fisher information	steepness of the score; curvature of the NLL	$I = E[-U'(\theta_0)]$
Score at the truth	heights of the curves at θ_0 ; normal by the CLT	$U(\theta_0) \approx \text{Normal}(0, I)$
Information identity	curves must move when the truth moves	$\text{Var}(U) = I$

Idea	Graphical meaning	Formula
Sampling distribution of the MLE	crossing points = height / slope	$\hat{\theta} \approx \text{Normal}(\theta_0, 1/I)$
Hessian (Wald) interval	width of the parabola at 1.92	$\hat{\theta} \pm 1.96/\sqrt{I}$
Likelihood ratio cutoff	squared standard normal	$2 \Delta \text{NLL} \sim \chi_1^2$, cutoff 1.92

One sentence holds it together: **a parabola-shaped log-likelihood and a straight-line score are the same assumption, and the normal sampling distribution, the Hessian interval, and the 1.92 cutoff all rest on it.**